

Отчёт о лабораторной работе

Лабораторная работа 4

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1 Цель работы

Приобретение практических навыков по установке и базовому конфигурированию HTTP-сервера Apache.

2 Выполнение лабораторной работы

Для начала запустим наш сервер через vagrant (рис. 2.1).

```
C:\work\nsandryushin\vagrant>vagrant up server
Bringing machine 'server' up with 'virtualbox' provider...
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: You assigned a static IP ending in ".1" or ":1" to this machine.
==> server: This is very often used by the router and can cause the
==> server: network to not work properly. If the network doesn't work
==> server: properly, try changing this IP.
==> server: Clearing any previously set forwarded ports...
==> server: Clearing any previously set network interfaces...
==> server: Preparing network interfaces based on configuration...
    server: Adapter 1: nat
    server: Adapter 2: intnet
==> server: Forwarding ports...
    server: 22 (guest) => 2222 (host) (adapter 1)
==> server: Running 'pre-boot' VM customizations...
==> server: Booting VM...
==> server: Waiting for machine to boot. This may take a few minutes...
    server: SSH address: 127.0.0.1:2222
    server: SSH username: vagrant
    server: SSH auth method: password
    server: Warning: Connection reset. Retrying...
The guest machine entered an invalid state while waiting for it
to boot. Valid states are 'starting, running, paused'. The machine is in the
'poweroff' state. Please verify everything is configured
properly and try again.
```

Рис. 2.1: Запуск сервера

Установим пакеты, необходимые для работы веб-сервера (рис. 2.2).

```

[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# LANG=C yum grouplist
Last metadata expiration check: 0:08:51 ago on Sat Sep 27 09:23:59 2025.
Available Environment Groups:
  Server
  Minimal Install
  Workstation
  KDE Plasma Workspaces
  KDE Plasma Mobile
  Custom Operating System
  Virtualization Host
Installed Environment Groups:
  Server with GUI
Installed Groups:
  Container Management
  Development Tools
  Headless Management
Available Groups:
  Desktop accessibility
  KDE Applications
  KDE
  KDE Multimedia support
  KDE Mobile
  KDE PIM
  KDE Software Development
  KDE Frameworks 6 Software Development
  Legacy UNIX Compatibility
  Smart Card Support
  Console Internet Tools
  .NET Development
  Graphical Administration Tools
  Network Servers
  RPM Development Tools
  Scientific Support
  Security Tools
  System Tools
[root@server.nsandryushin.net ~]# dnf -y groupinstall "Basic Web Server"
Last metadata expiration check: 0:09:25 ago on Sat 27 Sep 2025 09:23:59 AM UTC.
Dependencies resolved.
=====
Package                Architecture    Version                               Repository    Size
=====
Installing group/module packages:
httpd                  x86_64         2.4.63-1.el10_0.2                   appstream    52 k
httpd-manual           noarch         2.4.63-1.el10_0.2                   appstream    2.3 M
mod_fcgid              x86_64         2.3.9-35.el10                       appstream    72 k
mod_ssl               x86_64         1:2.4.63-1.el10_0.2                 appstream   108 k
Installing dependencies:
apr                   x86_64         1.7.5-2.el10                        appstream    128 k

```

Рис. 2.2: Установка пакетов

Проанализируем содержимое конфигурационных файлов сервера. Начнём с файла `/etc/httpd/conf/httpd.conf`. В нём содержатся основные настройки веб-сервера Apache. (рис. 2.3).

```

GNU nano 8.1                                httpd.conf
#
# This is the main Apache HTTP server configuration file. It contains the
# configuration directives that give the server its instructions.
# See <URL:http://httpd.apache.org/docs/2.4/> for detailed information.
# In particular, see
# <URL:http://httpd.apache.org/docs/2.4/mod/directives.html>
# for a discussion of each configuration directive.
#
# See the httpd.conf(5) man page for more information on this configuration,
# and httpd.service(8) on using and configuring the httpd service.
#
# Do NOT simply read the instructions in here without understanding
# what they do. They're here only as hints or reminders. If you are unsure
# consult the online docs. You have been warned.
#
# Configuration and logfile names: If the filenames you specify for many
# of the server's control files begin with "/" (or "drive:/" for Win32), the
# server will use that explicit path. If the filenames do *not* begin
# with "/", the value of ServerRoot is prepended -- so 'log/access_log'
# with ServerRoot set to '/www' will be interpreted by the
# server as '/www/log/access_log', where as '/log/access_log' will be
# interpreted as '/log/access_log'.
#
# ServerRoot: The top of the directory tree under which the server's
# configuration, error, and log files are kept.
#
# Do not add a slash at the end of the directory path. If you point
# ServerRoot at a non-local disk, be sure to specify a local disk on the
# Mutex directive, if file-based mutexes are used. If you wish to share the
# same ServerRoot for multiple httpd daemons, you will need to change at
# least PidFile.
#
ServerRoot "/etc/httpd"
#
# Listen: Allows you to bind Apache to specific IP addresses and/or
# ports, instead of the default. See also the <VirtualHost>
# directive.
#
# Change this to Listen on a specific IP address, but note that if
# httpd.service is enabled to run at boot time, the address may not be
# available when the service starts. See the httpd.service(8) man
# page for more information.
#
#Listen 12.34.56.78:80
Listen 80

```

Рис. 2.3: /etc/httpd/conf/httpd.conf

В файле /etc/httpd/conf/magic содержатся инструкции для определения MIME-типа файла по его содержимому, а не расширению (рис. 2.4).


```

GNU nano 8.1                                     magic
# Magic data for mod_mime_magic Apache module (originally for file(1) command)
# The module is described in /manual/mod/mod_mime_magic.html
#
# The format is 4-5 columns:
#   Column #1: byte number to begin checking from, ">" indicates continuation
#   Column #2: type of data to match
#   Column #3: contents of data to match
#   Column #4: MIME type of result
#   Column #5: MIME encoding of result (optional)

#-----
# Localstuff: file(1) magic for locally observed files
# Add any locally observed files here.

#-----
# end local stuff
#-----

#-----
# Java

0      short      0xcafe
>2     short      0xbabe      application/java

#-----
# audio: file(1) magic for sound formats
#
# from Jan Nicolai Langfeldt <janl@ifi.uio.no>,
#

# Sun/NeXT audio data
0      string      .snd
>12    belong      1      audio/basic
>12    belong      2      audio/basic
>12    belong      3      audio/basic
>12    belong      4      audio/basic
>12    belong      5      audio/basic
>12    belong      6      audio/basic
>12    belong      7      audio/basic

>12    belong      23     audio/x-adpcm

```

Рис. 2.4: /etc/httpd/conf/magic

В файле /etc/httpd/conf.d/autoindex.conf содержатся настройки для автоматического отображения списка файлов в директории (рис. 2.5).

```

GNU nano 8.1                                autoindex.conf
#
# Directives controlling the display of server-generated directory listings.
#
# Required modules: mod_authz_core, mod_authz_host,
#                   mod_autoindex, mod_alias
#
# To see the listing of a directory, the Options directive for the
# directory must include "Indexes", and the directory must not contain
# a file matching those listed in the DirectoryIndex directive.
#
#
# IndexOptions: Controls the appearance of server-generated directory
# listings.
#
IndexOptions FancyIndexing HTMLTable VersionSort

# We include the /icons/ alias for FancyIndexed directory listings. If
# you do not use FancyIndexing, you may comment this out.
#
Alias /icons/ "/usr/share/httpd/icons/"

<Directory "/usr/share/httpd/icons">
    Options Indexes MultiViews FollowSymLinks
    AllowOverride None
    Require all granted
</Directory>

#
# AddIcon* directives tell the server which icon to show for different
# files or filename extensions. These are only displayed for
# FancyIndexed directories.
#
AddIconByEncoding (CMP,/icons/compressed.gif) x-compress x-gzip

AddIconByType (TXT,/icons/text.gif) text/*
AddIconByType (IMG,/icons/image2.gif) image/*
AddIconByType (SND,/icons/sound2.gif) audio/*
AddIconByType (VID,/icons/movie.gif) video/*
AddIconByType /icons/bomb.gif application/x-coredump

```

Рис. 2.5: /etc/httpd/conf.d/autoindex.conf

В файле /etc/httpd/conf.d/manual.conf содержатся настройки для доступа к веб-странице с документацией Apache (рис. 2.6).

```

GNU nano 8.1                                manual.conf
#
# This configuration file allows the manual to be accessed at
# http://localhost/manual/
#
Alias /manual /usr/share/httpd/manual

<Directory "/usr/share/httpd/manual">
    Options Indexes
    AllowOverride None
    Require all granted

    RedirectMatch 301 ^/manual/(?:da|de|en|es|fr|ja|ko|pt-br|ru|tr|zh-cn)(/.*)$ "/manual$1"
</Directory>

```

Рис. 2.6: /etc/httpd/conf.d/manual.conf

В файле /etc/httpd/conf.d/userdir.conf содержатся настройки для доступа к публичным веб-директориям пользователей системы (рис. 2.7).

```

GNU nano 8.1                                userdir.conf
#
# UserDir: The name of the directory that is appended onto a user's home
# directory if a ~user request is received.
#
# The path to the end user account 'public_html' directory must be
# accessible to the webserver userid. This usually means that ~userid
# must have permissions of 711, ~userid/public_html must have permissions
# of 755, and documents contained therein must be world-readable.
# Otherwise, the client will only receive a "403 Forbidden" message.
#
<IfModule mod_userdir.c>
    #
    # UserDir is disabled by default since it can confirm the presence
    # of a username on the system (depending on home directory
    # permissions).
    #
    UserDir disabled

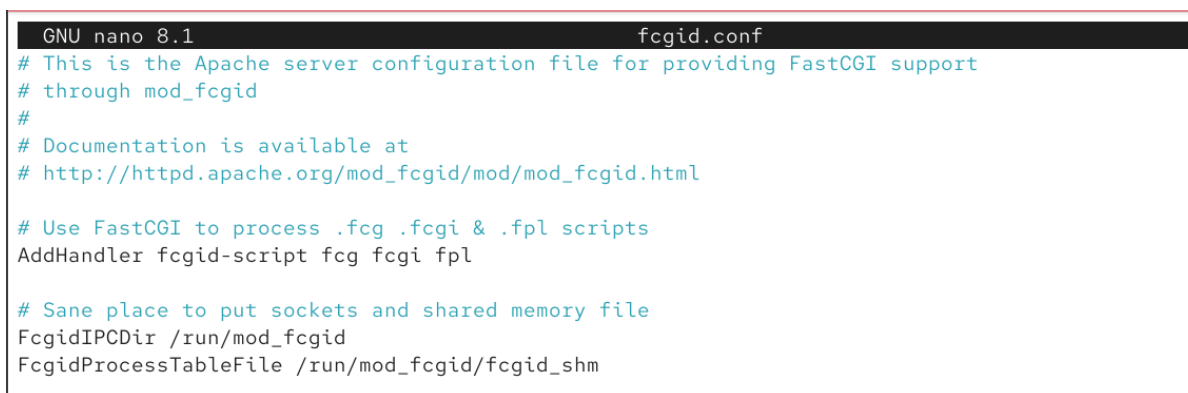
    #
    # To enable requests to ~/user/ to serve the user's public_html
    # directory, remove the "UserDir disabled" line above, and uncomment
    # the following line instead:
    #
    #UserDir public_html
</IfModule>

#
# Control access to UserDir directories. The following is an example
# for a site where these directories are restricted to read-only.
#
<Directory "/home/*/public_html">
    AllowOverride FileInfo AuthConfig Limit Indexes
    Options MultiViews Indexes SymLinksIfOwnerMatch IncludesNoExec
    Require method GET POST OPTIONS
</Directory>

```

Рис. 2.7: /etc/httpd/conf.d/userdir.conf

В файле `/etc/httpd/conf.d/fcgid.conf` содержатся настройки для модуля `mod_fcgid`, который используется для запуска скриптов (например, PHP) через FastCGI (рис. 2.8).

A screenshot of a terminal window showing the contents of the file `/etc/httpd/conf.d/fcgid.conf` using the GNU nano 8.1 editor. The file contains configuration directives for the mod_fcgid module, including comments about its purpose, documentation, and usage, as well as directives to add handlers and set paths for sockets and shared memory.

```
GNU nano 8.1                                fcgid.conf
# This is the Apache server configuration file for providing FastCGI support
# through mod_fcgid
#
# Documentation is available at
# http://httpd.apache.org/mod_fcgid/mod/mod_fcgid.html

# Use FastCGI to process .fcg .fcgi & .fpl scripts
AddHandler fcgid-script fcg fcgi fpl

# Sane place to put sockets and shared memory file
FcgidIPCDir /run/mod_fcgid
FcgidProcessTableFile /run/mod_fcgid/fcgid_shm
```

Рис. 2.8: `/etc/httpd/conf.d/fcgid.conf`

В файле `/etc/httpd/conf.d/ssl.conf` содержатся настройки для поддержки шифрования SSL/TLS (рис. 2.9).

```

GNU nano 8.1                                ssl.conf
#
# When we also provide SSL we have to listen to the
# standard HTTPS port in addition.
#
Listen 443 https

##
##  SSL Global Context
##
##  All SSL configuration in this context applies both to
##  the main server and all SSL-enabled virtual hosts.
##

#   Pass Phrase Dialog:
#   Configure the pass phrase gathering process.
#   The filtering dialog program ('builtin' is a internal
#   terminal dialog) has to provide the pass phrase on stdout.
SSLPassPhraseDialog exec:/usr/libexec/httpd-ssl-pass-dialog

#   Inter-Process Session Cache:
#   Configure the SSL Session Cache: First the mechanism
#   to use and second the expiring timeout (in seconds).
SSLSessionCache          shmcb:/run/httpd/sslcache(512000)
SSLSessionCacheTimeout   300

#
# Use "SSLCryptoDevice" to enable any supported hardware
# accelerators. Use "openssl engine -v" to list supported
# engine names. NOTE: If you enable an accelerator and the
# server does not start, consult the error logs and ensure
# your accelerator is functioning properly.
#
SSLCryptoDevice builtin
#SSLCryptoDevice ubsec

##
##  SSL Virtual Host Context
##

<VirtualHost _default_:443>

```

Рис. 2.9: /etc/httpd/conf.d/ssl.conf

В файле /etc/httpd/conf.d/welcome.conf содержатся настройки для отображения приветственной страницы по умолчанию после установки Apache (рис. 2.10).

```
GNU nano 8.1                               welcome.conf
#
# This configuration file enables the default "Welcome" page if there
# is no default index page present for the root URL. To disable the
# Welcome page, comment out all the lines below.
#
# NOTE: if this file is removed, it will be restored on upgrades.
#
<LocationMatch "^/+>"
    Options -Indexes
    ErrorDocument 403 /.noindex.html
</LocationMatch>

<Directory /usr/share/httpd/noindex>
    AllowOverride None
    Require all granted
</Directory>

Alias /.noindex.html /usr/share/httpd/noindex/index.html
Alias /poweredby.png /usr/share/httpd/icons/apache_pb3.png
Alias /system_noindex_logo.png /usr/share/httpd/icons/system_noindex_logo.png
```

Рис. 2.10: /etc/httpd/conf.d/welcome.conf

Теперь добавим службу http в фаервол для корректной работы (рис. 2.11).

```
[root@server.nsandryushin.net ~]# firewall-cmd --list-services
cockpit dhcp dhcpv6-client dns ssh
[root@server.nsandryushin.net ~]# firewall-cmd --get-services
0-AD RH-Satellite-6 RH-Satellite-6-capsule afp alvr amanda-client amanda-k5-client amqp amqps anno-1602 a
nno-1800 apcupsd aseqnet audit ausweisapp2 bacula bacula-client bareos-director bareos-filedaemon bareos-
storage bb bgp bitcoin bitcoin-rpc bitcoin-testnet bitcoin-testnet-rpc bittorrent-lsd ceph ceph-exporter
ceph-mon cfengine checkmk-agent civilization-iv civilization-v cockpit collectd condor-collector cratedb
ctdb dds dds-multicast dds-unicast dhcp dhcpv6 dhcpv6-client distcc dns dns-over-quic dns-over-tls docker
-registry docker-swarm dropbox-lansync elasticsearch etcd-client etcd-server factorio finger foreman fore
man-proxy freeipa-4 freeipa-ldap freeipa-ldaps freeipa-replication freeipa-trust ftp galera ganglia-clien
t ganglia-master git gpssd grafana gre high-availability http http3 https ident imap imaps iperf2 iperf3 i
pfs ipp ipp-client ipsec irc ircs iscsi-target isns jenkins kadmin kdeconnect kerberos kibana klogin kpas
swd kprop kshell kube-api kube-apiserver kube-control-plane kube-control-plane-secure kube-controller-man
ager kube-controller-manager-secure kube-nodeport-services kube-scheduler kube-scheduler-secure kube-work
er kubelet kubelet-readonly kubelet-worker ldap ldaps libvirt libvirt-tls lightning-network llmnr llmnr-c
lient llmnr-tcp llmnr-udp managesieve matrix mdns memcache minecraft minidlna mndp mongodb mosh mountd mp
d mqtt mqtt-tls ms-wbt mssql murmur mysql nbd nebula need-for-speed-most-wanted netbios-ns netdata-dashbo
ard nfs nfs3 nmea-0183 nrpe ntp nut opentelemetry openvpn ovirt-imageio ovirt-storageconsole ovirt-vmcons
ole plex pmcd pmproxy pmwebapi pmwebapis pop3 pop3s postgresql privoxy prometheus prometheus-node-exporte
r proxy-dhcp ps2link ps3netsrv ptp pulseaudio puppetmaster quassel radius radsec rdp redis redis-sentinel
rootd rpc-bind rquotad rsh rsyncd rtsp salt-master samba samba-client samba-dc sane settlers-history-col
lection sip sips slimevr slp smtp smtp-submission smtps snmp snmpv1 snmpv2 snmpv3 snmptrap spideroak-lans
ync spotify-sync squid ssdp ssh statsrv steam-lan-transfer steam-streaming stellaris stronghold-crusader
stun stuns submission supertuxkart svdrp svn syncthing syncthing-gui syncthing-relay synergy syscomlan sy
slog syslog-tls telnet tentacle terraria tftp tile38 tinc tor-socks transmission-client turn turns upnp-c
lient vdsd vnc-server vrrp warpinator wbem-http wbem-https wireguard ws-discovery ws-discovery-client ws-
discovery-host ws-discovery-tcp ws-discovery-udp wsdd wsdd-http wsman wsmans xdmcp xmpp-bosh xmpp-client
xmpp-local xmpp-server zabbix-agent zabbix-java-gateway zabbix-server zabbix-trapper zabbix-web-service z
ero-k zerotier
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=http
success
[root@server.nsandryushin.net ~]# firewall-cmd --add-service=http --permanent
success
```

Рис. 2.11: Настройка firewall

Запустим в отдельной вкладке journalctl с ключом -f для вывода информации логов в реальном времени (рис. 2.12).

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# journalctl -x -f
Sep 27 09:40:05 server.nsandryushin.net systemd[11194]: Started run-p14019-i14319.scope - [systemd-run] /
usr/bin/bash.
  Subject: A start job for unit UNIT has finished successfully
  Defined-By: systemd
  Support: https://wiki.rockylinux.org/rocky/support

  A start job for unit UNIT has finished successfully.

  The job identifier is 631.
Sep 27 09:40:09 server.nsandryushin.net PackageKit[13801]: daemon quit
Sep 27 09:40:09 server.nsandryushin.net systemd[1]: packagekit.service: Deactivated successfully.
  Subject: Unit succeeded
  Defined-By: systemd
  Support: https://wiki.rockylinux.org/rocky/support

  The unit packagekit.service has successfully entered the 'dead' state.
```

Рис. 2.12: journalctl

Теперь попробуем запустить службу httpd и включить в ней автозагрузку (рис. 2.13).

```
[root@server.nsandryushin.net ~]# systemctl enable httpd
Created symlink '/etc/systemd/system/multi-user.target.wants/httpd.service' → '/usr/lib/systemd/system/httpd.service'.
[root@server.nsandryushin.net ~]# systemctl start httpd
```

Рис. 2.13: запуск службы httpd

Вернёмся ко вкладке с journalctl. Как видим, запуск httpd был успешен (рис. 2.14).

```

Sep 27 09:40:41 server.nsandryushin.net systemd[1]: httpd-init.service: Deactivated successfully.
Subject: Unit succeeded
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit httpd-init.service has successfully entered the 'dead' state.
Sep 27 09:40:41 server.nsandryushin.net systemd[1]: Finished httpd-init.service - One-time temporary TLS
key generation for httpd.service.
Subject: A start job for unit httpd-init.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit httpd-init.service has finished successfully.

The job identifier is 11756.
Sep 27 09:40:41 server.nsandryushin.net systemd[1]: httpd-init.service: Consumed 4.758s CPU time, 2.8M me
mory peak.
Subject: Resources consumed by unit runtime
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

The unit httpd-init.service completed and consumed the indicated resources.
Sep 27 09:40:42 server.nsandryushin.net systemd[1]: Starting httpd.service - The Apache HTTP Server...
Subject: A start job for unit httpd.service has begun execution
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit httpd.service has begun execution.

The job identifier is 11610.
Sep 27 09:40:42 server.nsandryushin.net (httpd)[14238]: httpd.service: Referenced but unset environment v
ariable evaluates to an empty string: OPTIONS
Sep 27 09:40:42 server.nsandryushin.net systemd[1]: Started httpd.service - The Apache HTTP Server.
Subject: A start job for unit httpd.service has finished successfully
Defined-By: systemd
Support: https://wiki.rockylinux.org/rocky/support

A start job for unit httpd.service has finished successfully.

The job identifier is 11610.
Sep 27 09:40:42 server.nsandryushin.net httpd[14238]: Server configured, listening on: port 443, port 80

```

Рис. 2.14: Лог об успешном запуске

Теперь запустим клиент (рис. 2.15).


```

C:\work\nsandyushin\vagrant>vagrant up client
Bringing machine 'client' up with 'virtualbox' provider...
==> client: Clearing any previously set forwarded ports...
==> client: Fixed port collision for 22 => 2222. Now on port 2200.
==> client: Clearing any previously set network interfaces...
==> client: Preparing network interfaces based on configuration...
    client: Adapter 1: nat
    client: Adapter 2: intnet
==> client: Forwarding ports...
    client: 22 (guest) => 2200 (host) (adapter 1)
==> client: Running 'pre-boot' VM customizations...
==> client: Booting VM...
==> client: Waiting for machine to boot. This may take a few minutes...
    client: SSH address: 127.0.0.1:2200
    client: SSH username: vagrant
    client: SSH auth method: password
==> client: Machine booted and ready!
[client] GuestAdditions 7.1.4 running --- OK.
==> client: Checking for guest additions in VM...
==> client: Setting hostname...
==> client: Configuring and enabling network interfaces...
==> client: Mounting shared folders...
    client: C:/work/nsandyushin/vagrant => /vagrant
==> client: Machine already provisioned. Run `vagrant provision` or use the `--provision`
==> client: flag to force provisioning. Provisioners marked to run always will still run.
==> client: Running provisioner: common hostname (shell)...
    client: Running: C:/Users/mega_/AppData/Local/Temp/vagrant-shell120250927-18208-1fkfd4.sh
==> client: Running provisioner: client routing (shell)...
    client: Running: C:/Users/mega_/AppData/Local/Temp/vagrant-shell120250927-18208-jojyyc.sh
    client: Provisioning script /tmp/vagrant-shell

```

Рис. 2.15: Запуск клиента

Зайдём в клиенте в браузер и перейдём по адресу 192.168.1.1. Как видим, это страница, которая используется вебсервером по умолчанию. Таким образом, она даёт нам понять, что Служба httpd работает корректно, и клиент может получить доступ к содержимому веб-страницы (рис. 2.16).

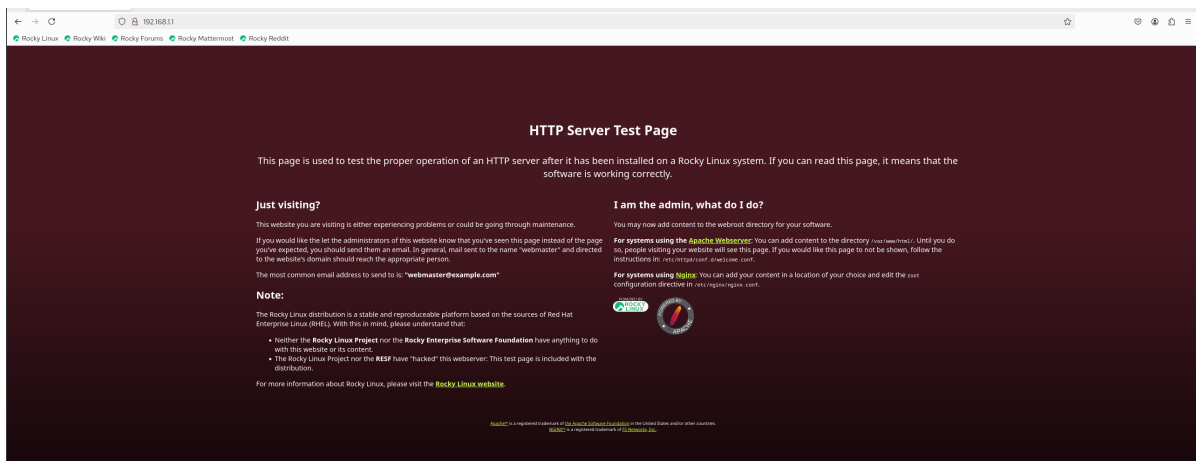


Рис. 2.16: Страница по умолчанию

Теперь выведем логи об ошибках веб-сервера в файле `/var/log/httpd/error_log` (рис. 2.17).

```
[root@server.nsandryushin.net ~]# tail -f /var/log/httpd/error_log
[Sat Sep 27 09:40:42.453011 2025] [suexec:notice] [pid 14238:tid 14238] AH01232: suEXEC mechanism enabled
(wrapper: /usr/sbin/suexec)
[Sat Sep 27 09:40:42.545059 2025] [lbmethod_heartbeat:notice] [pid 14238:tid 14238] AH02282: No slotmem f
rom mod_heartbeat
[Sat Sep 27 09:40:42.552802 2025] [systemd:notice] [pid 14238:tid 14238] SELinux policy enabled; httpd ru
nning as context system_u:system_r:httpd_t:s0
[Sat Sep 27 09:40:42.603145 2025] [mpm_event:notice] [pid 14238:tid 14238] AH00489: Apache/2.4.63 (Rocky
Linux) OpenSSL/3.2.2 mod_fcgid/2.3.9 configured -- resuming normal operations
[Sat Sep 27 09:40:42.603238 2025] [core:notice] [pid 14238:tid 14238] AH00094: Command line: '/usr/sbin/h
ttpd -D FOREGROUND'
[Sat Sep 27 09:47:29.922620 2025] [autoindex:error] [pid 14243:tid 14287] [client 192.168.1.30:42500] AH0
1276: Cannot serve directory /var/www/html/: No matching DirectoryIndex (index.html) found, and server-g
enerated directory index forbidden by Options directive
```

Рис. 2.17: `/var/log/httpd/error_log`

И выведем лог о доступе к веб-странице в файле `/var/log/httpd/access_log` (рис. 2.18).

```
[nsandryushin@server.nsandryushin.net ~]$ sudo -i
[sudo] password for nsandryushin:
[root@server.nsandryushin.net ~]# tail -f /var/log/httpd/access_log
192.168.1.30 - - [27/Sep/2025:09:47:29 +0000] "GET / HTTP/1.1" 403 7620 "-" "Mozilla/5.0 (X11; Linux x86_
64; rv:128.0) Gecko/20100101 Firefox/128.0"
192.168.1.30 - - [27/Sep/2025:09:47:30 +0000] "GET /icons/poweredby.png HTTP/1.1" 200 15443 "http://192.1
68.1.1/" "Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"
192.168.1.30 - - [27/Sep/2025:09:47:30 +0000] "GET /poweredby.png HTTP/1.1" 200 5714 "http://192.168.1.1/
" "Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"
192.168.1.30 - - [27/Sep/2025:09:47:30 +0000] "GET /favicon.ico HTTP/1.1" 404 196 "http://192.168.1.1/" "
Mozilla/5.0 (X11; Linux x86_64; rv:128.0) Gecko/20100101 Firefox/128.0"
^C
[root@server.nsandryushin.net ~]#
```

Рис. 2.18: `/var/log/httpd/access_log`

Остановим службу named, отвечающую за dns (рис. 2.19).

```
[root@server.nsandryushin.net ~]# systemctl stop named
```

Рис. 2.19: Остановка службы named

Поменяем файл зоны, добавив в него запись о том, что теперь у нас есть http сервер (www) (рис. 2.20).

```
GNU nano 8.1 /var/named/master/fz/nsandryushin.net Modified
$ORIGIN .
$TTL 86400      ; 1 day
nsandryushin.net IN SOA nsandryushin.net. server.nsandryushin.net. (
                        2025092002 ; serial
                        86400      ; refresh (1 day)
                        3600       ; retry (1 hour)
                        604800     ; expire (1 week)
                        10800      ; minimum (3 hours)
                        )
                        NS      nsandryushin.net.
                        A       192.168.1.1
                        AAAA    ::1
$ORIGIN nsandryushin.net.
$TTL 1200      ; 20 minutes
client        A       192.168.1.30
              DHCID   ( AAEBWH0317DS5G/2dhsaE/YY19eqf3aqET2vQfn6RTpz
                      JX4= ) ; 1 1 32
$TTL 86400     ; 1 day
dhcp          A       192.168.1.1
ns            A       192.168.1.1
server        A       192.168.1.1
www           A       192.168.1.1
```

Рис. 2.20: Файл зоны /var/named/master/fz/nsandryushin

То же самое сделаем в обратном файле зоны (рис. 2.21).

```

GNU nano 8.1 /var/named/master/rz/192.168.1 Modified
$ORIGIN .
$TTL 86400 ; 1 day
1.168.192.in-addr.arpa IN SOA 1.168.192.in-addr.arpa. server.nsandryushin.net. (
                                2025092002 ; serial
                                86400 ; refresh (1 day)
                                3600 ; retry (1 hour)
                                604800 ; expire (1 week)
                                10800 ; minimum (3 hours)
                                )
                                NS 1.168.192.in-addr.arpa.
                                A 192.168.1.1
                                PTR server.nsandryushin.net.
                                AAAA ::1
$ORIGIN 1.168.192.in-addr.arpa.
1 PTR server.nsandryushin.net.
PTR ns.nsandryushin.net.
PTR dhcp.nsandryushin.net.
1 PTR www.nsandryushin.net.
$TTL 1200 ; 20 minutes
30 PTR client.nsandryushin.net.
DHCID ( AAEBWH0317DS5G/2dhsaE/YY19eqf3aqET2vQfn6RTpz
JX4= ) ; 1 1 32

```

Рис. 2.21: Файл зоны /var/named/master/rz/192.168.1

Удалим файлы журналов из папок обратной и прямой зон (рис. 2.22).

```

[root@server.nsandryushin.net ~]# ls /var/named/master/fz/
nsandryushin.net nsandryushin.net.jnl
[root@server.nsandryushin.net ~]# ls /var/named/master/rz/
192.168.1 192.168.1.jnl
[root@server.nsandryushin.net ~]# rm /var/named/master/fz/nsandryushin.net.jnl /var/named/master/rz/192.1
68.1.jnl
rm: remove regular file '/var/named/master/fz/nsandryushin.net.jnl'? y
rm: remove regular file '/var/named/master/rz/192.168.1.jnl'? y
[root@server.nsandryushin.net ~]# ls /var/named/master/fz/
nsandryushin.net
[root@server.nsandryushin.net ~]# ls /var/named/master/rz/
192.168.1
[root@server.nsandryushin.net ~]#

```

Рис. 2.22: Удаление журналов

Теперь запустим dns службу и перейдём в папку с конфигурацией httpd (/etc/httpd/conf.d), создав там файлы конфигурации для двух страниц - server.nsandryushin.net и www.nsandryushin.net (рис. 2.23).

```

[root@server.nsandryushin.net ~]# systemctl start named
[root@server.nsandryushin.net ~]# cd /etc/httpd/conf.d
[root@server.nsandryushin.net conf.d]# touch server.nsandryushin.net.conf
[root@server.nsandryushin.net conf.d]# touch www.nsandryushin.net.conf
[root@server.nsandryushin.net conf.d]# nano server.nsandryushin.net.conf

```

Рис. 2.23: Создание конфигурационных файлов

Поместим следующее содержимое в первый из файлов (рис. 2.24).

```
GNU nano 8.1 server.nsandryushin.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@nsandryushin.net
DocumentRoot /var/www/html/server.nsandryushin.net
ServerName server.nsandryushin.net
ErrorLog logs/server.nsandryushin.net-error_log
CustomLog logs/server.nsandryushin.net-access_log common
</VirtualHost>
```

Рис. 2.24: server.nsandryushin.net

И следующее содержимое во второй файл (рис. 2.25).

```
GNU nano 8.1 www.nsandryushin.net.conf
<VirtualHost *:80>
ServerAdmin webmaster@nsandryushin.net
DocumentRoot /var/www/html/www.nsandryushin.net
ServerName www.nsandryushin.net
ErrorLog logs/www.nsandryushin.net-error_log
CustomLog logs/www.nsandryushin.net-access_log common
</VirtualHost>
```

Рис. 2.25: www.nsandryushin.net

В папке /var/www/html создадим папки server.nsandryushin.net и www.nsandryushin.net. В каждой из них создадим файл index.html, в каждый из которых запишем простую приветственную фразу (рис. 2.26).

```
[root@server.nsandryushin.net html]# cd /var/www/html
[root@server.nsandryushin.net html]# mkdir server.nsandryushin.net
[root@server.nsandryushin.net html]# d /var/www/html/server.nsandryushin.net/
bash: d: command not found...
[root@server.nsandryushin.net html]# cd /var/www/html/server.nsandryushin.net/
[root@server.nsandryushin.net server.nsandryushin.net]# touch index.html
[root@server.nsandryushin.net server.nsandryushin.net]# echo "Welcome to the server.nsandryushin.net server." > index.html
[root@server.nsandryushin.net server.nsandryushin.net]# cd /var/www/html
[root@server.nsandryushin.net html]# mkdir www.nsandryushin.net
[root@server.nsandryushin.net html]# cd /var/www/html/www.nsandryushin.net/
[root@server.nsandryushin.net www.nsandryushin.net]# touch index.html
[root@server.nsandryushin.net www.nsandryushin.net]# echo "Welcome to the www.nsandryushin.net server." > index.html
```

Рис. 2.26: Файлы index.html

Теперь обновим метки для SELinux и перезапустим службу httpd (рис. 2.27).

```
[root@server.nsandryushin.net www.nsandryushin.net]# chown -R apache:apache /var/www
[root@server.nsandryushin.net www.nsandryushin.net]# restorecon -vR /etc
Relabeled /etc/NetworkManager/system-connections/eth1.nmconnection from unconfined_u:object_r:user_tmp_t:s0
to unconfined_u:object_r:NetworkManager_etc_rw_t:s0
[root@server.nsandryushin.net www.nsandryushin.net]# restorecon -vR /var/named
[root@server.nsandryushin.net www.nsandryushin.net]# restorecon -vR /var/www
[root@server.nsandryushin.net www.nsandryushin.net]# systemctl restart httpd
[root@server.nsandryushin.net www.nsandryushin.net]# █
```

Рис. 2.27: Обновление меток и перезапуск службы

Теперь с клиента попробуем перейти по адресу `server.nsandryushin.net`. Как видим, нам вывело страницу с той самой фразой, которую мы записывали в `index.html` ранее (рис. 2.28).

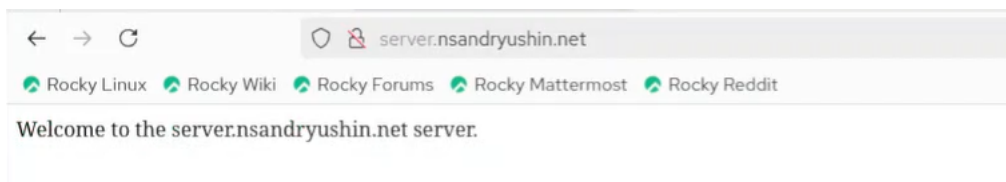


Рис. 2.28: `server.nsandryushin.net`

Аналогичную ситуацию мы видим с сайтом по адресу `www.nsandryushin.net` (рис. 2.29).

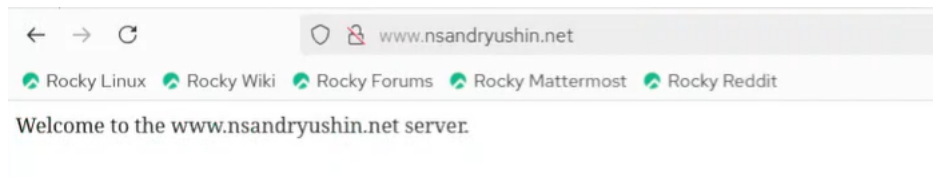


Рис. 2.29: `www.nsandryushin.net`

Теперь обновим конфигурацию `vagrant`, поместив обновлённые конфиги и все созданные в ходе лабораторной работы файлы в папку `/vagrant/provision/server`. Кроме того, создадим скрипт `http.sh` (рис. 2.30).

```
[root@server.nsandryushin.net www.nsandryushin.net]# cd /vagrant/provision/server
[root@server.nsandryushin.net server]# mkdir -p /vagrant/provision/server/http/etc/httpd/conf.d
[root@server.nsandryushin.net server]# mkdir -p /vagrant/provision/server/http/var/www/html
[root@server.nsandryushin.net server]# cp -R /etc/httpd/conf.d/* /vagrant/provision/server/http/etc/httpd/
/conf.d/
[root@server.nsandryushin.net server]# cp -R /var/www/html/* /vagrant/provision/server/http/var/www/html
[root@server.nsandryushin.net server]# cd /vagrant/provision/server/dns/
[root@server.nsandryushin.net dns]# cp -R /var/named/* /vagrant/provision/server/dns/var/named/
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run-20250920'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/data/named.run'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind.jnl'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/dynamic/managed-keys.bind'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/fz/nsandryushin.net'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/master/rz/192.168.1'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.ca'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.empty'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.localhost'? y
cp: overwrite '/vagrant/provision/server/dns/var/named/named.loopback'? y
[root@server.nsandryushin.net dns]# cd /vagrant/provision/server
[root@server.nsandryushin.net server]# touch http.sh
[root@server.nsandryushin.net server]# chmod +x http.sh
[root@server.nsandryushin.net server]#
```

Рис. 2.30: Обновление конфигурации vagrant

В скрипт `http.sh` пропишем следующее содержимое, позволяющее настроить наш http сервер (рис. 2.31).

```
GNU nano 8.1 http.sh
#!/bin/bash
echo "Provisioning script $0"
echo "Install needed packages"
dnf -y groupinstall "Basic Web Server"
echo "Copy configuration files"
cp -R /vagrant/provision/server/http/etc/httpd/* /etc/httpd
cp -R /vagrant/provision/server/http/var/www/* /var/www
chown -R apache:apache /var/www
restorecon -vR /etc
restorecon -vR /var/www
echo "Configure firewall"
firewall-cmd --add-service=http
firewall-cmd --add-service=http --permanent
echo "Start http service"
systemctl enable httpd
systemctl start httpd
```

Рис. 2.31: Скрипт `http.sh`

Отредактируем `Vagrantfile`, добавив в его конфигурацию запуск созданного ранее скрипта (рис. 2.32).


```
server.vm.provision "server dummy",
    type: "shell",
    preserve_order: true,
    path: "provision/server/01-dummy.sh"

server.vm.provision "server dns",
    type: "shell",
    preserve_order: true,
    path: "provision/server/dns.sh"

server.vm.provision "server dhcp",
    type: "shell",
    preserve_order: true,
    path: "provision/server/dhcp.sh"

server.vm.provision "server http",
    type: "shell",
    preserve_order: true,
    path: "provision/server/http.sh"
```

Рис. 2.32: Vagrantfile

3 Выводы

В результате выполнения лабораторной работы были получены навыки работы и настройки http сервера